Martin Thoma

Curriculum Vitæ

Bavaria, Germany
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Work Experience

2021

Lead Developer
Building Django Backends for Cashlink

Cashlink is a Blockchain/FinTech Startup with an incredible team. I'm improving the overall code quality and implementing new features. The environment is highly asynchronous.

2017 - 2021

IT Consultant for Netlight Consulting GmbH
Building ML applications and Data Science tasks

I've built a dynamic pricing system using reinforcement learning, a recommendation engine at a video streaming service, helped to build and establish a local and offshore data science team at an SDAX company. The analytical tasks involved A/B tests, churn analysis, fraud prediction.
My work typically was done with Python, sklearn / NumPy / SciPy, Keras (Tensorflow backend), XGBoost, Jupyter Notebooks, Pandas, pytest, Flask and nginx. Usually in teams using Scrum or Kanban and Jira / Confluence. Often Packaging, Docker, AWS S3, ECR / ECS, DynamoDB, Cloudwatch. Sometimes MySQL, PostgreSQL, or SQLite. Of course, using git. Occassionally, Kafka, GraphQL, building REST APIs and Selenium were involved. Recently, I worked on security topics (AppSec, SDLC, SCA, SAST, DAST, threat modeling).

2014

Student research assistant
Developing neural nets for handwriting recognition

My bachelors thesis includes getting on-line data of handwritten mathematical symbols, preprocessing, extracting features and using neural nets to classify those symbols. The data was collected with write-math.com. All results are available there, too.

2013 - 2015

Software Developer
Building a continuous learning architecture for KIT

I've implemented and integrated an unsupervised acoustic model training framework into KIT lecture translator system for automatic model adaption.

2012 - 2013

Tutor for programming & Scientific lector
Teaching, LaTeX, Computer Science

Teaching students coding conventions and basic OOP in Java; improving a script for computer engineering. → presentations

2011

Freelancer at KTC
Programming for a consulting company

At KTC, I gained first experiences with a business rule engine and a big, but algorithmically not challenging project. It was my first contact with Consulting and Java, working for DAX companies.

Education

2014 – 2017 **Master of Science** (Computer Science)
Thesis about **Analysis and Optimization of Convolutional Neural Network Architectures** (**Link**)
Karlsruhe Institute of Technology

2011 – 2014 **Bachelor of Science** (Computer Science)
Karlsruhe Institute of Technology (KIT)
Carnegie Mellon University (CMU)
Thesis about **On-line Recognition of Handwritten Mathematical Symbols** (**Link**)

Awards

2013 **German Academic Scholarship Foundation**
Permanent Membership was granted

2010 **Winner Bundeswettbewerb Informatik**
Federal Competition for Computer Science

2008 **1st prize**
data analysis competition at University of Augsburg

2007 **Prize for science and research**
FOCUS pupils competition

Technological Stack

Basics	JavaScript, jQuery, Linux, SQL, AWS (S3, EC2, ECR / ECS, Lambda, Sagemaker), PHP, GraphQL, Django
Intermediate	LaTeX, Java, HTML, CSS, SQLAlchemy Docker, git, CI/CD pipelines
Expert	Python and it's eco-system, including packaging, pytest, pre-commit, Flask, Numpy, SciPy, Pandas, Jupyter, OOP, Type Annotations, Pypika, REST, …

Language Skills

German	mother tongue
English	Cambridge Certificate – C1

Work Experience

2011

Student research assistant at Institute of Toxicology and Genetics, KIT

participating in a university research project

In summer 2011 I worked on a research project at KIT. I have written bash scripts for file conversions and re-written a slow Mathematica script in a much faster Python version. It was my first contact with big data, clustering algorithms, data centers using Hadoop, C++, VTK, cmake

2011 - 2012

Building my own Startup

MATHOS - a calendar app for schools

I have started to work as a freelancer at the beginning of 2011. I have developed an online-service which helped schools to coordinate their dates. I have sold this online service to two schools in bavaria and three other schools were interested. Unfortunately, the ministry of education of Bavaria released an application with similar functionality in 2012. This was the reason why I decided to shut down my service.

2010

Thesis about the RSA cryptosystem

Explaining the foundations of asymmetric cryptography

Understanding asymmetric cryptograph with the RSA algorithm was the first time I went down the rabbithole of university-level mathematics. I could combine my enthusiasm for computer science, algorithms and mathematics in this first academic work. → read more

2006 - 2011

Algorithm Engineering and Ethical Hacking

ProjectEuler and many more

I like solving logical, algorithmical or math puzzles and participated in competitions. It started with trying to break into systems where the creators purposely left a security hole such as a possible SQL injection. From those HackIt's, I continued to solve programming and algorithm engineering puzzles in 2006 and I still like them. This was the reason why I participated in a practical curse at KIT for preparation for ICPC. It was fun, but I found out that many people are much faster in producing C++ code that passed the tests than I am. However, as I've been very successfull at the Federal Competition for Computer Science ("Bundeswettbewerb Informatik") it seems as if I'm better in problem solving if I get more time to think about it.

About me

I love learning new stuff and giving that knowledge to others, often in the form of blog posts. Besides my studies, I have built a machine learning students group called *Machine Learning Karlsruhe*. In this group we organized regular „Paper Discussion Groups" in which we talked about papers about convolutional networks.
I like playing board games, reading SciFi novels, to go hiking, bouldering, and climbing.

Private Projects

09/2020	**Flake8-simplify** *Automatic code simplification* → read more
02/2016	**The WiLI benchmark dataset** *written language identification* → read more
02/2016	**A Survey of Semantic Segmentation** *writing a review paper about the work in the area* → read more
01/2016	**Creativity in Machine Learning** *a little project to help people understand my fascination about the topic* → read more
05/2015	**Semantic Segmentation with CNNs** *classifying street for self-driving cars*
11/2013	**Book about Geometry and Topology** *writing an introduction to geometry and topology* → read more
06/2013	**Interpolation** *creating an interactive HTML5/JS-example for interpolation* → read more
06/2012	**Matrix multiplication** *examining algorithms and libraries for matrix multiplication* → read more
since 09/2011	**Blogging on martin-thoma.com** *about Algorithms, the Web, University, ...*
06/2011	**Community Chess** *This is a platform for programmers. They can use the API to create A.I.s that play chess agains each other.* → read more

Online Courses

09/2013	**Artificial Intelligence for Robotics** *finished 10/2013*	Udacity
06/2013	**Introduction to Artificial Intelligence** *finished 08/2013*	Udacity
05/2012	**Algorithms I** *finished 07/2012*	Stanford
06/2010	**Introduction to Computer Science and Programming** *finished 09/2010*	MIT

Profiles

StackExchange	careers.stackoverflow.com/thoma
arXiv	arxiv.org/a/thoma_m_1
LinkedIn	linkedin.com/in/martin-thoma